

Technical Datasheet 4G Gateway with 4DI, 4DO, & 4AI

Product Overview: -

The 4G Gateway is an advanced networking device designed to provide seamless connectivity and data management capabilities for various applications. With support for 4G LTE connectivity, cloud connectivity via MQTT protocol, and robust storage options, it offers a comprehensive solution for remote monitoring and control systems. Equipped with an integrated I/O interface featuring **4 Digital Inputs (DI)**, **4 Digital Outputs (DO)**, and **4 Analog Inputs (AI)**, alongside a Modbus RS485 port and micro SIM compatibility, it caters to a wide range of industrial and IoT applications. The intuitive web-based UI interface allows for easy configuration and management, making it an ideal choice for both professionals and enthusiasts.



Specifications

General:-

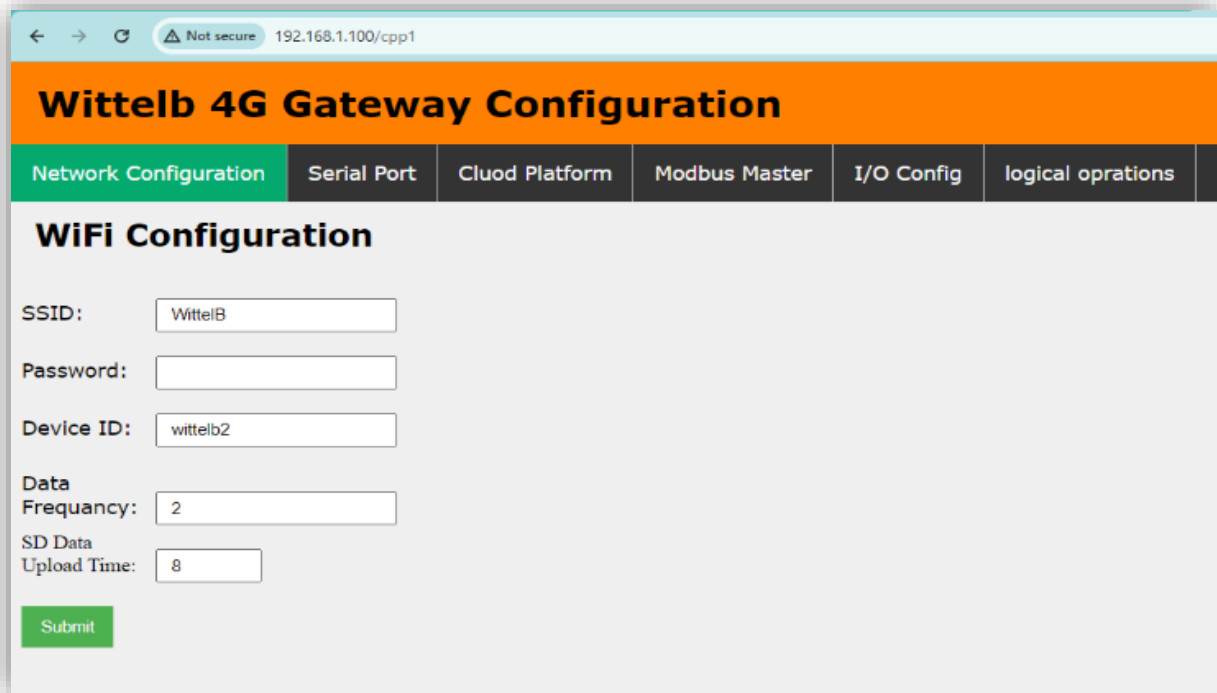
- **4G LTE Connectivity:** Utilizes 4G LTE technology for reliable and high-speed internet access. If 4G is not available, the system automatically switches to Wi-Fi. If both 4G and Wi-Fi are unavailable, the data is securely stored in the SD card for later upload.
- **Cloud Connectivity (MQTT):** Seamlessly connects to cloud platforms using the MQTT protocol for efficient data transmission and management.
- **Event Storage:** Offers on board event storage, capable of storing up to 50,000 events or more. Includes an external memory card slot for easy data retrieval in the event of 4G network issues.
- **Modbus RS485 Port:** Features 1 port of Modbus RS485 for integration with industrial control systems and sensors.
- **GPS Location Data:** Provides GPS location data for monitoring of remote equipment.
- **Micro Sim Compatibility:** Supports micro-Sim card for flexible and convenient connectivity option.
- **Web-based UI Interface:** Intuitive web-based user interface for easy configuration, monitoring, and management.
- **Dimensions:** 110 mm L x 110 mm B x 50 mm H.
- **Power:** Input Power – 15-40 VDC, Typical – 12V DC @ 150mA.
- **Operating Temperature:** 0 – 60C (32 ~ 140F).
- **Storage Temperature:** 20 - 70C (-4 ~ 158F).
- **Storage Humidity:** 5 ~ 95 % RH, non – Condensing.

Configuration Settings: -



- Power up the module by connecting a 12-24 V DC supply.
- Insert the SIM card into the designated slot.
- Place the Micro SD card into its designated slot.
- Connect MODBUS RTU Multiple Slave devices (up to 31) to the A & B Terminal Of Module.
- Access the module's configuration hotspot (ID: Wittelb-4G-GATEWAY, PASS: 123456789).
- Default IP Address of Module is "192.168.1.100/cpp1".
- Upon entering the provided IP address, a configuration webpage will be accessible.

Page1:-



← → 🔍 Not secure 192.168.1.100/cpp1

Wittelb 4G Gateway Configuration

Network Configuration	Serial Port	Cloud Platform	Modbus Master	I/O Config	logical oprations
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WiFi Configuration

SSID:

Password:

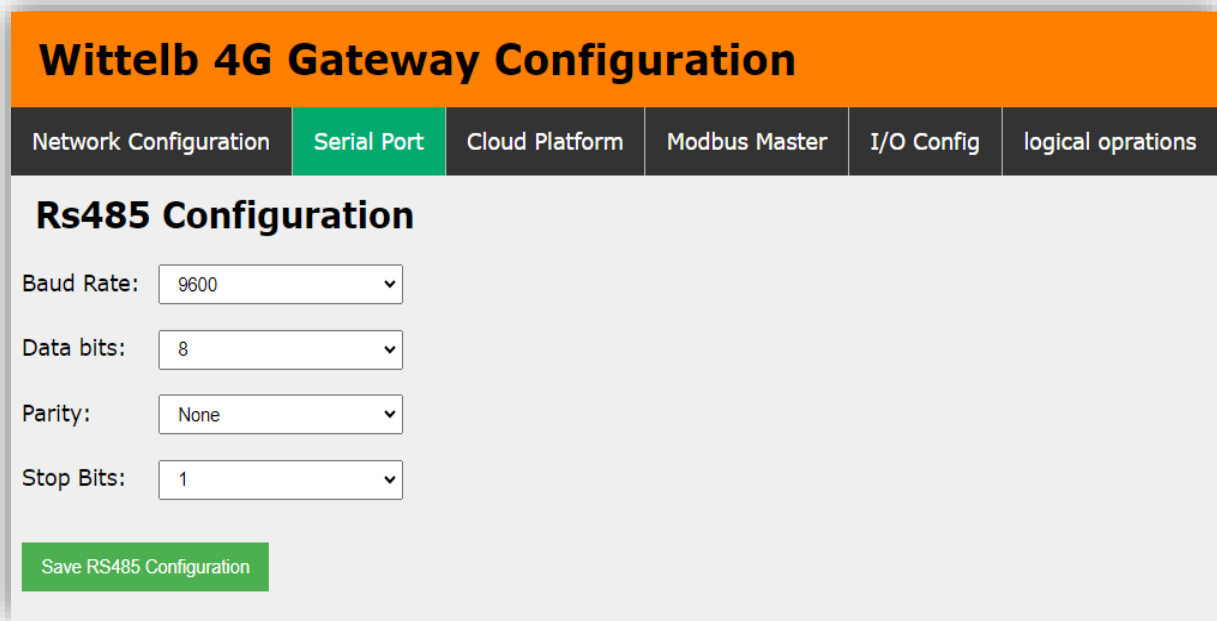
Device ID:

Data Frequency:

SD Data Upload Time:

- Enter Device ID (Should be unique for every device).
- Enter data frequency in min to receive data on cloud.
- Enter time to upload SD card data to cloud.

Page2:- On second page you can set device Baud Rate (9600 / 19200 / 38400 / 57600 / 115200), Data Bits (5 / 6 / 7 / 8), Parity (None / Odd / Even), and Stop Bits (1 / 2) for RS485 Communication



Wittelb 4G Gateway Configuration

Network Configuration	Serial Port	Cloud Platform	Modbus Master	I/O Config	logical oprations
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Rs485 Configuration

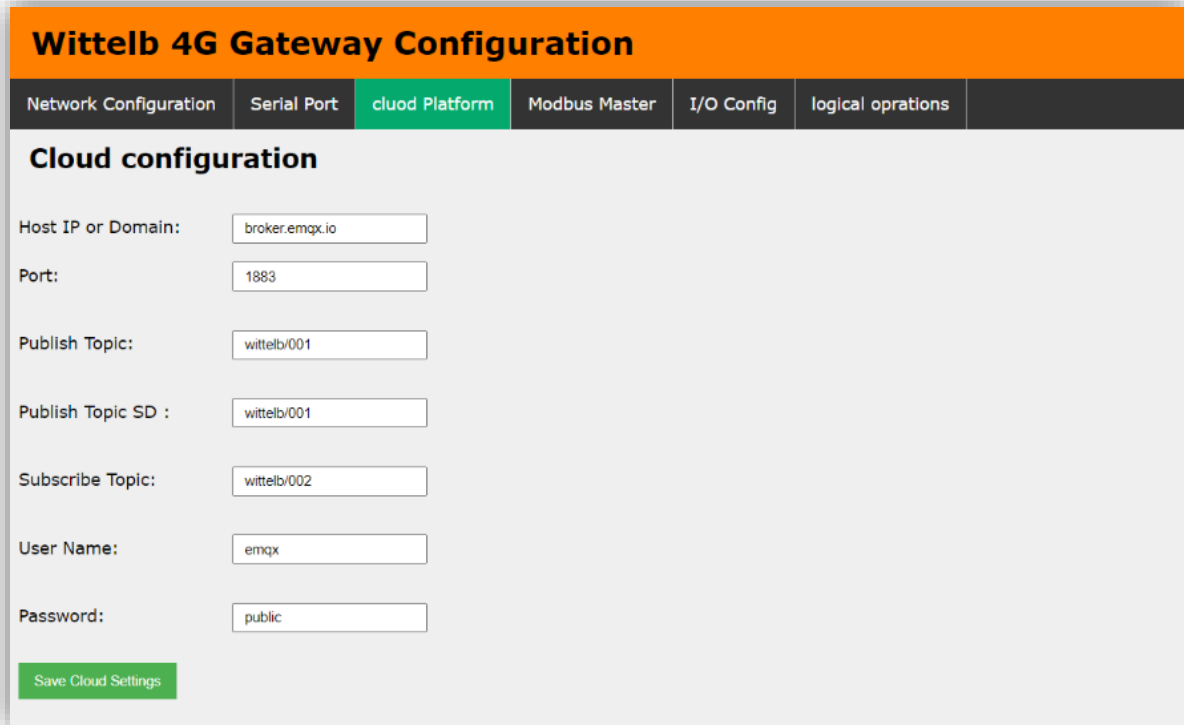
Baud Rate:

Data bits:

Parity:

Stop Bits:

Page3:-



Wittelb 4G Gateway Configuration

Network Configuration	Serial Port	Cloud Platform	Modbus Master	I/O Config	logical operations
Cloud configuration					
Host IP or Domain:		broker.emqx.io			
Port:		1883			
Publish Topic:		wittelb/001			
Publish Topic SD :		wittelb/001			
Subscribe Topic:		wittelb/002			
User Name:		emqx			
Password:		public			
Save Cloud Settings					

- Navigate to page 3 on the cloud platform.
- Input the Host IP or Domain.
- Enter the port Number.
- Enter the topics for publishing and receiving messages.
- Provide a username and password if required.
- Allow a few seconds for the module to restart and establish a connection to MQTT.

Page4:-

- Go to page 4 on the Modbus Master interface.
- Input the name and slave address for the designated slave.
- Specify the register address from which you want to read slave data.
- Indicate the register end address for the slave.
- Enter the function code corresponding to the desired slave.

Wittelb 4G Gateway Configuration

Network Configuration | Serial Port | Cloud Platform | **Modbus Master** | I/O Config | logical operations | About

Name	Slave Address	Register start	Quantity	Function Code	Actions
MFM	5	0	50	3	<button>Delete</button>

Add Row Submit

Page5:-

Wittelb 4G Gateway Configuration

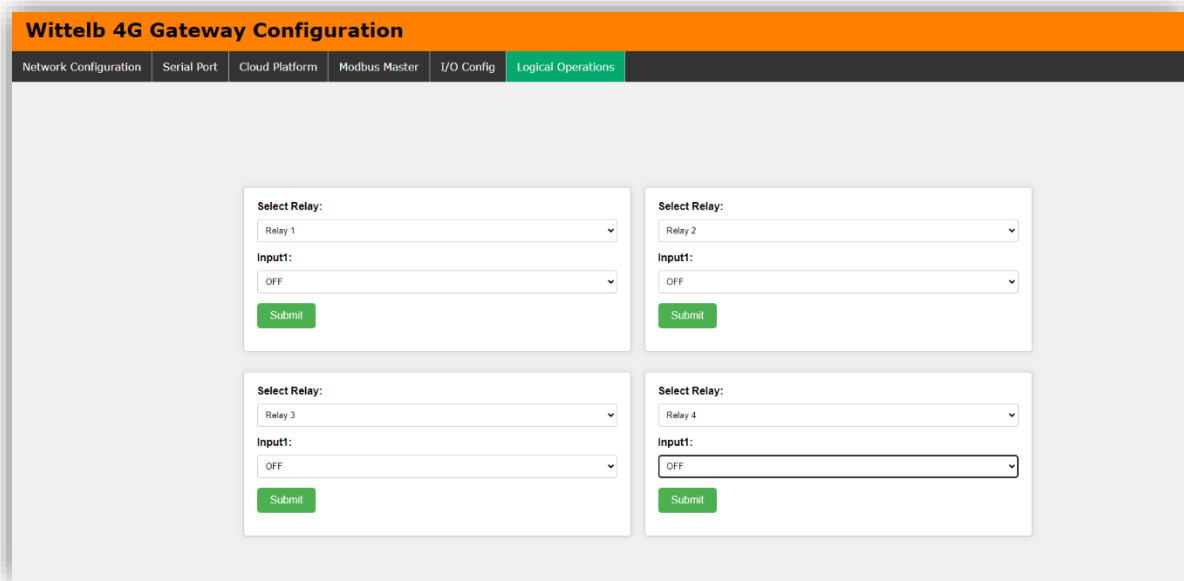
Network Configuration | Serial Port | Cloud Platform | Modbus Master | **I/O Config** | logical operations

Sl.No	Input/Output	Name	Value
1	DO1	RL11	
2	DO2	RL12	
3	DO3	RL13	1
4	DO4	RL14	
5	DI1	DI01	1
6	DI2	DI02	
7	DI3	DI03	
8	DI4	DI04	
9	AI1	Ai10-0	259
10	AI2	Ai10-1	197
11	AI3	Ai10-2	213
12	AI4	Ai10-3	227
13	AI1	Ai12-4	552
14	AI2	Ai12-5	2
15	AI3	Ai12-6	
16	AI4	Ai12-7	1

Submit

- Go to page 5 on the I/O Config interface.
- You can view real-time values of both analog and digital inputs on the baseboard.
- Additionally, you have the option to assign specific names to each input/output (I/O) as needed.

Page6:-

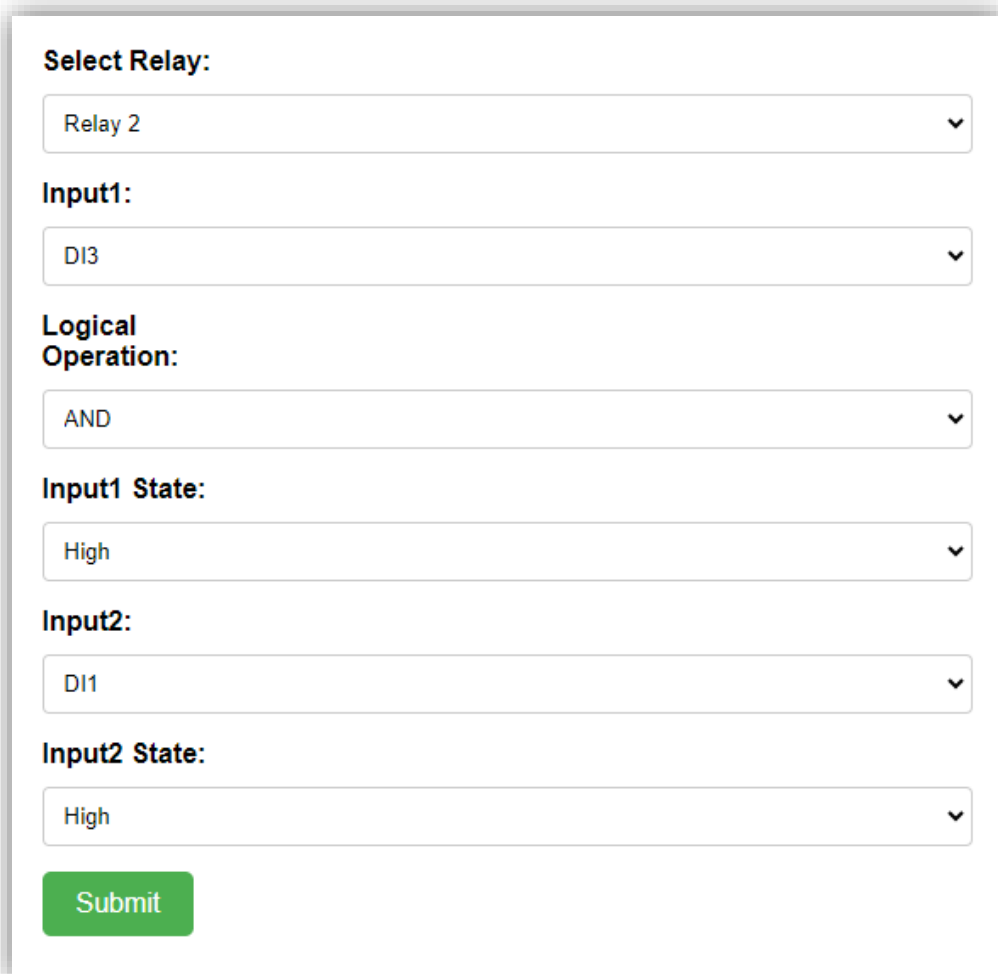


Wittelb 4G Gateway Configuration

Network Configuration | Serial Port | Cloud Platform | Modbus Master | I/O Config | **Logical Operations**

The interface displays four configuration panels for relays 1 through 4. Each panel includes a 'Select Relay' dropdown menu, an 'Input1' dropdown menu, and a 'Submit' button. In the screenshot, all 'Select Relay' and 'Input1' dropdowns are set to 'OFF'.

- Navigate to page 6 on the Logical Operations interface.



Select Relay:

Relay 2

Input1:

DI3

Logical Operation:

AND

Input1 State:

High

Input2:

DI1

Input2 State:

High

Submit

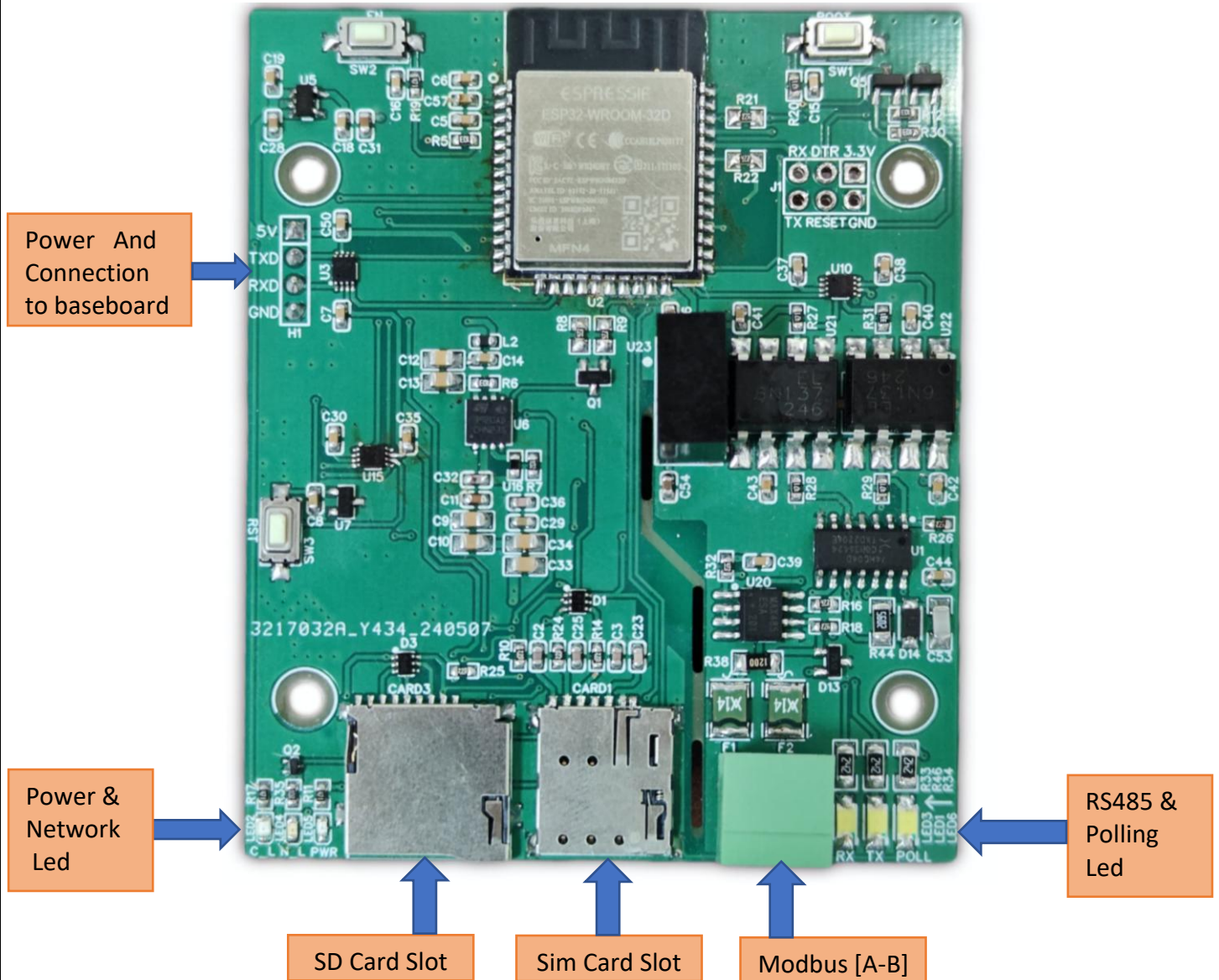
- Here, you can configure logical AND and OR operations for each of the 4 relays.
- You have the flexibility to assign any two digital inputs to a relay and choose the desired logical operation.
- Additionally, you can assign an analog input to a relay and define minimum and maximum threshold values for relay operation.
- If you prefer not to use any logical operation, simply select the "OFF" option for input one.

Input & Output module Configuration Settings: -

- The Gateway has 4 Digital Outputs (DO) with Relay of 10 Amps 230V,
- 4 Analog Inputs (AI) to connect with 4-20ma or 0-10 v sensors,
- 4 Digital Inputs (DI) will help monitor the status with logic voltage input between 4-40V DC. Additionally, we have also provided 12 V voltage ref and GND pins for DI detection logic.
- Digital outputs are normally open.
- Data of Digital Inputs (DI) and Analog Inputs (AI) would be directly collected by the Processor and transmitted over 4G to the Cloud based on the Poling frequency.
- Configure each Analog Input (AI) to operate with either 5V or 10V and choose between 10-bit and 12-bit resolution by adjusting the jumper settings.
- You can trigger DO from cloud by sending Following Message

Relay No	ON	OFF
DO1	"DO1":1	"DO1":1
DO2	"DO2":1	"DO1":1
DO3	"DO3":1	"DO1":1
DO4	"DO4":1	"DO1":1

4G Gateway connection Diagram:



Additional Features: -

Communication ports are isolated

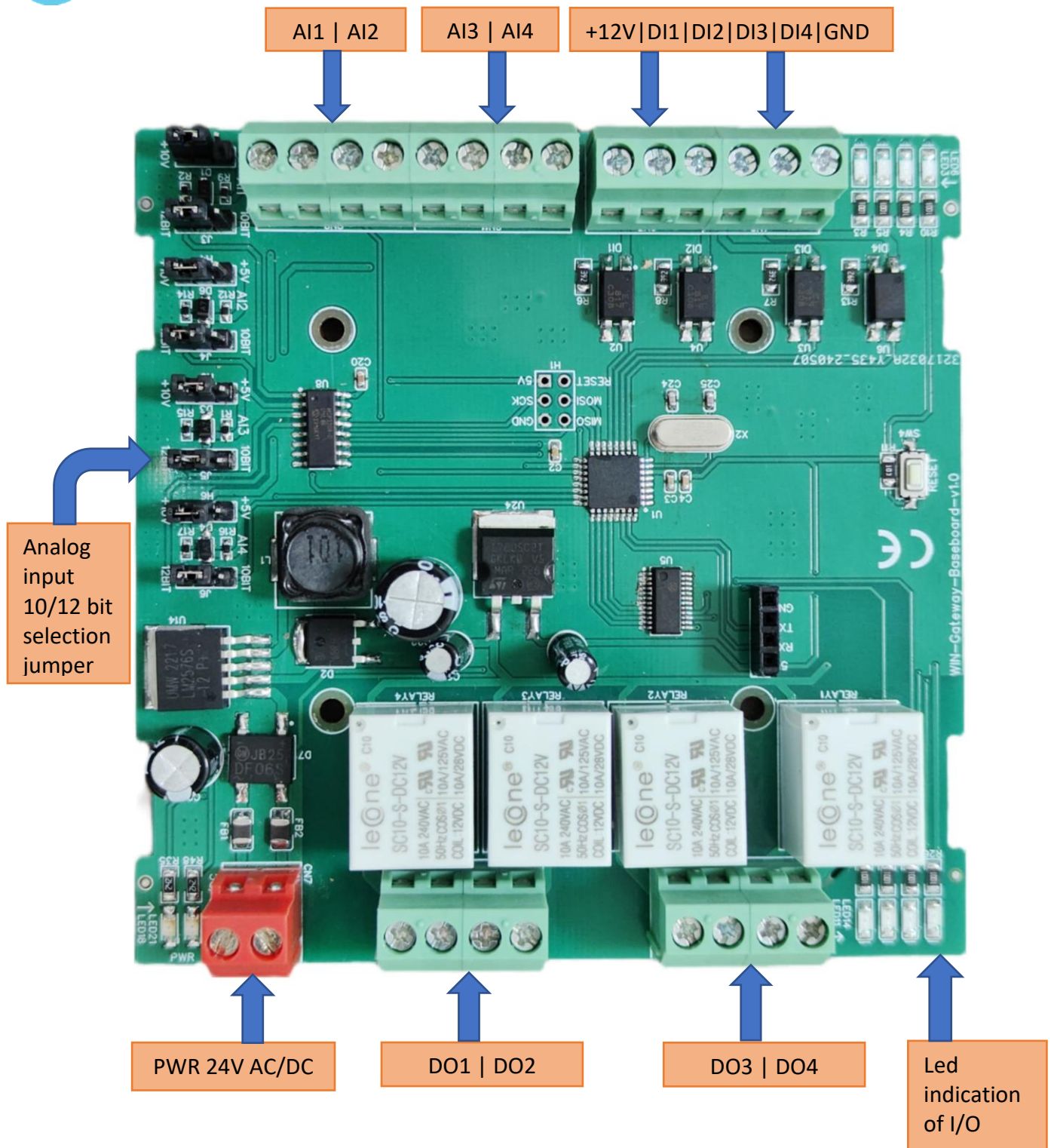
Input power reverse polarity safety

ESD Safety IEC 61000-4-2, $\pm 30\text{KV}$ contact, $\pm 30\text{KV}$ air

EFT IEC 61000-4-4, 50A (5/50ms)

400V isolation.

WIN-GW-4G-M-IO-A



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